



Solar power

1 1250 watts is equal to... (Tick 1 correct answer)

- ☐ 0.125 kilowatts
- ☒ 1.25 kilowatts
- ☐ 12.5 kilowatts
- ☐ 125 000 kilowatts
- ☐ 1 250 000 kilowatts

2 A house has a set of 5 solar panels. Each panel produces 1.08 kWh of electricity per day. In total, the set of 5 panels produce 5.4 kWh of electricity per day. (Fill in the blank)

3 A home owner spends £10 200 on solar panels. The panels save £1275 per year on electricity bills. How many years will it take for the annual savings to cover the cost of the solar panels?

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4 In ideal conditions, a solar panel would generate 1.4 kWh of electricity per day. However, its performance is reduced so that it produces 30% less electricity. Therefore, it produces 0.98 kWh per day. (Fill in the blank)

5 The average cost of electricity could be calculated as: 24.5 p per kWh consumed + 60 p daily standing charge. For a house that uses 8 kWh of electricity per day, the cost would be £ 2.56 per day. (Fill in the blank)



- 6 The formula calculates the amount of electricity produced by a solar panel. A solar panel has a power rating of 450 watts. With 5 hours of peak sunlight, the panel produces 2.25 kWh of electricity. (Fill in the blank)

$$E = PH$$

E = electricity produced (kWh)

P = power rating of panel (kW)

H = amount of peak sunlight (hrs)

